Lesson 10: Understanding Psychological Measurement

English Stage 4 (Psychology) · Unit 2: Research Ethics and Measurement

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 4.1 Prelude to Psychological Measurement; 4.2 Understanding Psychological Measurement.

In this lesson

You learn what measurement means in psychology, why constructs such as self-esteem cannot be observed directly, how researchers move from a conceptual definition to an operational definition, and the four levels of measurement. You practise writing formal definitions and using nominalisation, two features of academic English that run through this section.

Learning goals

After this lesson you should be able to:

- Define measurement, psychological construct, conceptual definition and operational definition.
- Classify measures as self-report, behavioral or physiological.
- Identify the level of measurement of a variable and say what it allows you to conclude.
- Write formal definitions and turn verb phrases into noun phrases.
- Write conceptual and operational definitions for the variables in your proposal.

Key language

Term or phrase	Meaning	Example
measurement	The assignment of scores to individuals so that the scores represent some characteristic of the individuals.	Measurement requires some systematic procedure.
psychometrics	The measurement of psychological variables and constructs.	Psychological measurement is often referred to as psychometrics.
construct	A variable that cannot be observed directly because it represents a tendency to behave in certain ways or a complex pattern of behavior and internal processes.	personality traits, emotional states, attitudes, abilities
conceptual definition	A description of a construct in terms of the behaviors and internal processes involved, and how it relates to other variables.	a conceptual definition of neuroticism
operational definition	A definition of a variable or construct in terms of precisely how it will be measured.	the number of aggressive acts in 20 minutes
self-report measure	participants report on their own thoughts, feelings and actions	the Rosenberg Self-Esteem Scale

Term or phrase	Meaning	Example
behavioral measure	some other aspect of participants' behavior is observed and recorded	the backward digit span task
physiological measure	A measure that involves recording a physiological variable.	heart rate, cortisol, brain activity
converging operations	Multiple operational definitions of the same construct; when they produce the same pattern of results, the construct is being measured effectively.	The measures of stress are all correlated with each other.
levels of measurement	Four ways of assigning scores (nominal, ordinal, interval, ratio) that give increasing amounts of quantitative information.	Stevens called them "scales of measurement".
reverse scoring	scoring an item in the opposite direction, because agreement shows less of the construct	Reverse the scoring for Items 3, 5, 8, 9 and 10.
be referred to as	be called (formal)	Psychological measurement is often referred to as psychometrics.
in terms of	using the idea of; with reference to	a definition of a variable in terms of precisely how it is to be measured

Rules and explanations

The ideas

Why measurement matters (4.1). MacDonald and Martineau (2002) found that a negative mood increased female students' intentions to have unprotected sex, but only for students low in self-esteem. They had to measure three things: self-esteem, mood (to check that thinking negative thoughts really produced a negative mood) and intentions. For self-esteem they used the **Rosenberg Self-Esteem Scale**: 10 items rated *Strongly Agree*, *Agree*, *Disagree* or *Strongly Disagree*. Items 1, 2, 4, 6 and 7 score 3, 2, 1, 0; Items 3, 5, 8, 9 and 10 are reverse-scored (0, 1, 2, 3). The total is the sum, from 0 to 30.

What is measurement? Any **systematic procedure** for assigning scores that represent a characteristic: a bathroom scale, a meat thermometer, or in psychology the **backward digit span task** (the longest list of digits a person can repeat in reverse order = working memory capacity) or the **Beck Depression Inventory** (the sum of 21 self-report ratings of symptoms over the past 2 weeks = current level of depression). No particular instrument is required.

Constructs. Age and height are easy to measure; intelligence and self-esteem are not. Such **constructs** cannot be observed directly because they are *tendencies* (a highly extraverted student may be reading quietly right now) or involve *internal processes* (fear involves the nervous system, thoughts and feelings). Each is a summary of many behaviors and processes. The **Big Five** are five broad personality dimensions, each divided into six "facets"; the book's figure lists them, and PSYC 110

Lecture 13 gives them as OCEAN: openness, conscientiousness, extroversion, agreeableness and neuroticism.

Conceptual definitions. Neuroticism is "people's tendency to experience negative emotions such as anxiety, anger, and sadness across a variety of situations"; the definition may add that it has a strong genetic component, is fairly stable and correlates with physical symptoms. Researchers do not use dictionaries: many constructs have no everyday word (working memory capacity), and researchers need more precise definitions, which they test and revise.

Operational definitions say precisely how a variable is measured:

Type	What happens	Book's examples
Self-report	participants report their own thoughts, feelings and actions	Rosenberg scale; Beck Depression Inventory
Behavioral	other behavior is observed and recorded, in the lab or in natural settings	backward digit span; Bandura's Bobo doll study: the number of hits, punches and kicks a child gave the doll in 20 minutes
Physiological	physiological processes are recorded	heart rate, blood pressure, galvanic skin response, hormones, brain activity

Every construct has several operational definitions. **Stress** can be measured with the Social Readjustment Rating Scale (life events in the past year, with points: divorce 73, changed jobs 36, change in sleeping habits 16, total 125), the Hassles and Uplifts Scale (everyday stressors), the Perceived Stress Scale (feelings of stress), blood pressure or cortisol. Using several is **converging operations**: because these measures correlate with each other and with immune functioning, researchers can conclude that "stress is negatively correlated with immune system functioning".

Levels of measurement (Stevens, 1946):

Level	Information	Book's examples	Central tendency
Nominal	category labels only, no order	marital status, ethnicity, favourite colour	mode
Ordinal	rank order; intervals not necessarily equal	finishing place in a race; "very dissatisfied" to "very satisfied"	median or mode
Interval	equal intervals, no true zero	Fahrenheit and Celsius; IQ	mean, median or mode
Ratio	equal intervals and a true zero	height, weight, number of siblings, Kelvin, money	mean, median or mode; ratios are meaningful

So 80 degrees Fahrenheit is not "twice as hot" as 40, and an IQ of 140 is not twice an IQ of 70, but someone with six siblings has twice as many as someone with three. The levels show how general measurement is (categorizing and ranking are measurement too) and guide the choice of statistics.

Language focus

1. Formal definitions. The section defines terms with the pattern *term* + *is/are* + *class word* + *defining clause*:

- "**Measurement is** the assignment of scores to individuals **so that** the scores represent some characteristic of the individuals."
- "**An operational definition is** a definition of a variable **in terms of** precisely how it is to be measured."
- "**Self-report measures are those in which** participants report on their own thoughts, feelings, and actions." It also names a category after describing it: "These kinds of variables **are called** constructs"; "Psychological measurement **is often referred to as** psychometrics."

2. Nominalisation. Academic English often turns verbs into nouns, so that a process can be the subject of a sentence: *assign scores* becomes "**the assignment of** scores"; *measure the variables* becomes "one of the primary ones is **the measurement of** the relevant variables"; *observe behavior* becomes "**the observation of** people's behavior". Note the pattern *the* + noun + *of* + object.

> **Note for Arabic speakers** - In general definitions, English uses no article with uncountable abstract nouns: "Measurement is ...", not "The measurement is ...", although Arabic uses the definite article here. - A nominalisation needs *of* before its object: "the assignment of scores", not "the assignment scores" or "the assigning scores".

Practice

1. Reading questions (sections 4.1 and 4.2)

1. What three variables did MacDonald and Martineau have to measure, and why did they measure mood?
2. On the Rosenberg scale, a person answers *Strongly Agree* to Item 1, *Agree* to Item 3 and *Strongly Disagree* to Item 10. How many points does each answer get? What are the lowest and highest possible totals?
3. Give the book's two reasons why constructs cannot be observed directly.
4. Why do researchers not simply look up constructs in the dictionary?
5. What was Bandura's operational definition of physical aggression?
6. On the Social Readjustment Rating Scale, what is the score of a person who only changed jobs and had a change in sleeping habits in the past year?
7. Why are converging operations useful?
8. Why can we not say that a person with an IQ of 140 is twice as intelligent as a person with an IQ of 70?

2. Classifying measures (sections 4.1 and 4.2 have no exercises; the chapter's exercises are in Lesson 12) A. Self-report, behavioral or physiological? 1. the Beck Depression Inventory 2. the backward digit span task 3. counting a child's acts of aggression 4. cortisol levels 5. the Perceived Stress Scale 6. galvanic skin response B. Which level of measurement? 7. marital status 8. finishing

place in a 100-m race 9. a runner's time in seconds 10. temperature in degrees Celsius 11. number of siblings 12. satisfaction from "very dissatisfied" to "very satisfied" 13. temperature in kelvins

3. Language A. Write a formal definition (term + class word + defining clause) from the section's description.

1. working memory capacity ("their ability to hold in mind and think about several pieces of information all at the same time")
2. converging operations B. Nominalise the words in bold, starting with *The*.
3. **Measuring** the relevant variables was one of the primary challenges.
4. Researchers **observe how children behave** in natural settings. (Begin: "The observation of ...")
5. Scores **are assigned** to individuals **systematically**. (Begin: "The systematic ...")

Speak with your tutor

- "Let me explain the difference between a construct, a conceptual definition and an operational definition, using neuroticism. Correct my explanation and my definitions."
- "In PSYC 110 Lecture 13, Professor Bloom names the Big Five as OCEAN. Help me suggest a self-report and a behavioral measure of extroversion."
- "In PSYC 110 Lecture 8, Professor Bloom reads eleven numbers and asks students how many they remembered. What is the operational definition there, and how is it different from the backward digit span task?"
- "In PSYC 110 Lecture 17, Professor Bloom says that sleep is hard to study because you cannot ask people what is happening while they sleep, so researchers use the EEG. Discuss self-report and physiological measures of sleep for my project with me."

Writing task

Write the **definitions of variables** for your proposal (sleep and memory in university students, or your own topic): 200-300 words. For each main variable (in the default project, nap versus wake after learning, and memory):

- a conceptual definition (1-2 sentences);
- a precise operational definition, saying exactly what is counted or rated, and when (for example: "the number of word pairs, out of 40, that participants correctly recall after a nap opportunity of up to 90 minutes or the same time resting quietly while awake");
- the type of measure (self-report, behavioral or physiological) and its level of measurement, with a reason;
- a second operational definition that could provide converging operations.

Use formal definition patterns and at least two nominalisations. If you work on your own topic, define your own variables in the same way.

Self-editing checklist:

- Does each conceptual definition describe behaviors or processes, not just give a synonym?
- Is each operational definition precise enough for another researcher to repeat?
- Did I name the type of measure and the level of measurement correctly?
- Did I give a second, different operational definition?
- Did I use *the* … *of* correctly in my nominalisations?
- Did I avoid *The* before general uncountable nouns in definitions?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "Levels of Measurement" (WikiEducator), the page the book drew on for nominal, ordinal, interval and ratio scales: http://wikieducator.org/Introduction_to_Research_Methods_In_Psychology/Theories_and_Measurement/Levels_of_Measurement

Answer key

1. Reading questions

1. Self-esteem, mood and intentions to have unprotected sex. Mood was measured to be sure that the attempt to create a negative mood (thinking negative thoughts) had worked (4.1).
2. Item 1: 3 points; Item 3 (reverse-scored): 1 point; Item 10 (reverse-scored): 3 points (7 points for these three items). The total can range from 0 to 30 (10 items, 0 to 3 points each).
3. Constructs often represent tendencies to think, feel or act in certain ways, not what the person is doing now; and they often involve internal processes that are not obvious to an outside observer.
4. Many constructs have no counterpart in everyday language, and researchers need definitions that are more detailed and precise and describe the world more accurately; they test and revise them.
5. The number of specifically defined acts of physical aggression (hitting the Bobo doll with a mallet, punching it, kicking it) that a child committed during a 20-minute period, counted from film.
6. 36 + 16 = 52 points.
7. When scores from different operational definitions are closely related and show similar patterns, this is good evidence that the construct is measured effectively and is useful, and it allows general conclusions instead of narrow, measure-specific ones.
8. IQ is measured at the interval level: it has no true zero point (a score of 0 would not mean no IQ), so ratios of scores are not meaningful.

2. Classifying measures A. 1. self-report 2. behavioral 3. behavioral 4. physiological 5. self-report 6. physiological B. 7. nominal 8. ordinal 9. ratio (not classified in the book, but 0 seconds means no time, and 12 s is twice 6 s) 10. interval 11. ratio 12. ordinal 13. ratio

3. Language A. (example answers) 1. Working memory capacity is a person's ability to hold in mind and think about several pieces of information at the same time. 2. Converging operations are the use of multiple operational definitions of the same construct, either within a study or across studies. B. 3. The measurement of the relevant variables was one of the primary challenges. 4. The observation of children's behavior in natural settings … (for example: "… is a behavioral measure"). 5. The systematic assignment of scores to individuals … (for example: "… is the essence of measurement").

My notes